
Computer Security and Privacy

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These notes cover the core vocabulary and reasoning framework of computer security: adversaries, threat models, security policies, mechanisms, and how to argue that a system is secure.

CHAPTER 1

Lecture 1 : Basics

Why a course on security?

Ordinary engineering asks for **correctness**, **safety** and **robustness** against accidental errors. Security additionally requires that properties hold in the presence of a **deliberate, strategic adversary**.

1.1 Computer security

Computer security

Properties of a computer system must hold in the presence of a **resourced strategic adversary**.

1.1.1 Security properties

CIA triad

- **Confidentiality** — prevention of unauthorized *disclosure* of information.
- **Integrity** — prevention of unauthorized *modification* of information.
- **Availability** — prevention of unauthorized *denial of service*.

CIA on a bank account

The adversary should not be able to *read* my statement (confidentiality), *change* my balance (integrity), or *block* my access to the account (availability).

Beyond the CIA triad

Modern systems also care about **authenticity**, **anonymity**, **isolation**, **non-repudiation**, and more.

1.2 Security policy

Security policy

A high-level description of the **security properties** that must hold, in relation to **assets** and **principals**.

- **Assets (objects)** — anything of value that needs protection (data, files, memory, ...).
- **Principals (subjects)** — people, programs, services, ... (never the adversary).

Properties in terms of principals and assets

Confidentiality	prevent unauthorized disclosure	authorized users may read a file
Integrity	prevent unauthorized modification	authorized programs may write a file
Availability	prevent unauthorized denial of service	authorized services can access a file

1.3 The adversary

Threat model

Describes the **resources** available to the adversary and its **capabilities** (observe, influence, corrupt, ...).

Adversary

A malicious entity aiming to breach the security policy. The adversary is **strategic**: it always picks the **optimal** way to use its resources to violate a security property.

Key vocabulary

- **Vulnerability** — a specific weakness an adversary could exploit (*e.g. a password shown in plaintext*).
- **Threat** — the feared event, i.e. the adversary's goal (*e.g. a hacker wants to steal money from the bank*).
- **Harm** — what actually happens if the threat materializes (*e.g. the adversary steals the money*).

If the properties hold within the threat model, no vulnerability can be exploited to realize a threat, so no harm occurs.

1.4 Security mechanisms

Security mechanism

A technical mechanism (software, hardware, cryptography, procedures, ...) used to ensure that the security policy is not violated by an adversary within the threat model.

Policy → mechanism

Policy: the transaction log must not be tampered with by a single employee.

Mechanism: replicate the log across multiple machines so that no employee has access to all copies.

Composing mechanisms

Security does **not** increase linearly with the number of mechanisms.

- **Defence in depth** (AND) — secure as long as *one* mechanism holds.
- **Weakest link** (OR) — insecure as soon as *any one* mechanism fails.

Humans are part of the system

Social engineering, phishing, and weak/reused/written-down passwords are frequently the weakest link — but that is no excuse to neglect the rest of the system.

1.5 Showing a system is secure

Attacker vs. defender

Attacker: one way to violate one property is enough (within the threat model).

Defender: must show **no** adversary strategy violates the policy — security is always relative to a **specific threat model**.

Secure

A system is **secure** if an adversary constrained by a specific threat model cannot violate the security policy.

Security argument

A rigorous argument (verbal or mathematical) that the security mechanisms in place are effective in maintaining the security policy — **subject to the assumptions of the threat model**.

Composition is hard

A useful threat model must constrain the adversary, otherwise no security argument is possible. Security arguments for the **composition** of mechanisms are notoriously hard to get right.

1.6 Security engineering process

Three-stage process

1. **High-level specification** — architecture, security policy (principals, assets, properties), threat model.
2. **Security design** — select/design mechanisms; state the security argument (which mechanism maintains which property).
3. **Secure implementation** — implement the mechanisms, ensure conformance to the design, perform security testing.

Summary

- Security problems always involve an **adversary**, who is **strategic**.
- The adversary's capabilities define a **threat model**.
- **Security mechanisms** aim at fulfilling a **security policy within a threat model**.
- Showing security means providing a **security argument**.